

How Can Clinical and Demographic Factors Be Used to Identify Patients at Risk of Heart Disease?

Milestone 3 – Initial Results and Model Evaluation

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Chapter 1 – Introduction

Cardiovascular disease is one of the major causes of mortality globally, and continues to exert a considerable strain on healthcare systems. Coronary heart disease (CHD) affects many people, and risk factors can be recognized years before the condition causes symptoms. Early identification of these patients gives healthcare practitioners the opportunity to prescribe lifestyle adjustment, preventative therapy, and routine surveillance which may lower the chance of future cardiovascular events.

Developments in data science and machine learning have offered new potential to help clinical decision-making by examining massive healthcare datasets and uncovering complicated correlations between patient attributes and illness outcomes. Unlike traditional statistical approaches, machine learning algorithms are able to understand the patterns from prior patient data and make predictions for future patients. These models may be useful decision support tools for health care workers, but they are not designed to replace clinical judgment.

This research uses the Framingham Heart Study dataset to study if demographic, behavioral and clinical characteristics may be utilized to predict the 10-year risk of coronary heart disease. The dataset includes patient variables like age, gender, smoking status, blood pressure, cholesterol levels, glucose levels, body mass index, diabetes status and other cardiovascular risk factors. The outcome variable TenYearCHD is a binary variable coded as 1 if a person got coronary heart disease in a 10-year follow-up period, and 0 otherwise.

In Milestone 1, the project focused on defining the research challenge, selecting a suitable dataset, and building a project plan. At Milestone 2 this work was extended by performing exploratory data analysis, looking at the data quality, missing values, class imbalance and designing the machine learning pipeline. Milestone 3 is to create the pipeline, preprocess the dataset, train various machine learning models, evaluate the performance and interpret the findings based on these previous milestones.

Prior to the creation of the model, many preprocessing procedures were employed. Missing values were imputed with the median, numerical variables were standardized with feature scaling, Recursive Feature Elimination (RFE) was performed to select the most informative predictors and the Synthetic Minority Oversampling Technique (SMOTE) was implemented to balance the training data. Preprocessing techniques were carried out to enhance model performance and to mitigate the effect of missing data and class imbalance.

In this work we assessed three supervised machine learning algorithms, Logistic Regression, Random Forest and Extreme Gradient Boosting (XGBoost). Different assessment criteria such as Accuracy, Precision, Recall, F1-score and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) were used to compare the prediction performance. Since the project is about a healthcare problem of prediction, special emphasis was given to Recall and ROC-AUC, since it is usually more critical to identify patients at risk accurately rather than to have the greatest accuracy score.

In order to increase the transparency of the predictive models, SHAPley Additive explanations (SHAP) were also employed to determine which factors most significantly contributed to the model predictions. This helps to understand how specific patient characteristics impact the prediction of coronary heart disease and gives more insight into the clinical value of the selected parameters.

The general purpose of this milestone is not only to evaluate the predicted performance of different machine learning algorithms, but also to provide a comprehensive and reproducible machine learning workflow from raw data preparation to interpretable model outcomes. The results of this work can be useful for future decision support systems that can help health care workers in identifying the persons with higher risk of getting coronary heart disease.

Chapter 2 – Project Objectives

For this purpose, the following objectives were established:

- Perform descriptive statistics and exploratory data analysis to investigate and understand the properties of the Framingham Heart Study dataset.
- To discover and fix data quality concerns including missing values, outliers, and class imbalance before model building.
- To preprocess the dataset with suitable techniques such as median imputation, feature scaling, Recursive Feature Elimination (RFE) and SMOTE to enhance the quality of the training data.
- Build and compare three supervised machine learning models: Logistic Regression, Random Forest and XGBoost.
- Evaluate model performance using Accuracy, Precision, Recall, F1-score and ROC-AUC, and conclude whether the model is more suitable for predicting risk of coronary heart disease.
- SHAP may be used to understand the model predictions and determine the most important clinical and demographic parameters related to coronary heart disease.
- To design a reproducible machine learning pipeline including data preparation, model construction, assessment and interpretation, using acceptable data science principles.

Chapter 3 – Research Questions

The following research questions guided the analysis and model development throughout this project.

Research Question 1

Can demographic and clinical variables be used to accurately predict the ten-year risk of coronary heart disease?

Research Question 2

Which machine learning model provides the best predictive performance for identifying individuals at risk of coronary heart disease?

Research Question 3

Which demographic and clinical factors have the greatest influence on predicting coronary heart disease?

Chapter 4 – Dataset Description

The dataset utilized in this study is the Framingham Heart Study dataset, one of the most popular public datasets for cardiovascular disease prediction. The Framingham Heart Study began in 1948 in Framingham, Massachusetts, USA, to determine variables that contribute to the development of cardiovascular disease over time. It has now become one of the most prominent long-term medical research initiatives and has done a lot to increase our understanding of the risk factors for heart disease.

The dataset utilized for this experiment is the version with data acquired from adult participants in standard clinical evaluations. Each row represents one person, and each column indicates a demographic, behavioral or clinical trait assessed during the research.

The outcome variable, TenYearCHD, is whether the subject got coronary heart disease within ten years of the baseline exam. This allows the dataset to be used for a supervised binary classification task to determine if a person is in the low-risk category or the high-risk group.

4.2 Characteristics of the dataset

The data set has 4,240 patient records and 16 variables. There are 15 predictor variables and one target variable. The predictor variables include demographic information, lifestyle habits, medical history and clinical data related to cardiovascular health.

The dataset contains both numerical and category variables. Continuous variables included age, total cholesterol, systolic blood pressure, diastolic blood pressure, body mass index (BMI), heart rate, glucose level, and smoking per day. Binary categorical factors include sex, smoking status, diabetes status, hypertension, prior stroke, and usage of blood pressure medication.

The dataset contains a mixture of information about the patient and so, it is a proper dataset for training and evaluating different machine learning classification models for heart disease prediction.

4.3 Variables Studied

The machine learning models were created utilizing the following predictor variables:

S.No	Variable	Description
1	Male	Sex of the participant (0 = Female, 1 = Male)
2	Age	Age of the participant in years
3	Education	Education level
4	Current Smoker	Current smoking status
5	Cigs Per Day	Average number of cigarettes smoked per day
6	BP Meds	Whether the participant was taking blood pressure medication
7	Prevalent Stroke	Previous history of stroke
8	Prevalent Hypertension	History of hypertension
9	Diabetes	Diabetes status
10	Total Cholesterol	Total cholesterol level
11	Sys BP	Systolic blood pressure
12	Dia BP	Diastolic blood pressure
13	BMI	Body Mass Index
14	Heart Rate	Resting heart rate
15	Glucose	Blood glucose level

16	TenYearCHD	Target variable: Indicates whether the participant developed coronary heart disease within 10 years (0 = No, 1 = Yes).
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4.4 Quality of Data

Prior to model creation, the data set was screened for any data quality concerns. The exploratory data analysis revealed missing values in numerous variables such as education, cigsPerDay, BPMeds, totChol, BMI, heartRate and glucose. No duplicate records were found in the study.

There were also some outliers in several numerical variables. Instead, such numbers were saved as they are true clinical measures that may occur in real patients. Instead, we have used suitable preprocessing approaches, including median imputation, feature scaling and robust model assessment, to decrease their impact on model performance.

Another key feature of the dataset was the unbalance between the two target classes. Some 85% of the patients had not got coronary heart disease and just 15% had the condition after ten years. To mitigate this imbalance, SMOTE was used solely on the training set following the train-test split. Thus, we were able to increase the model's capacity to detect high risk patients without data leaking.

4.5 Reasons for Choosing This Dataset

The dataset utilized in this study was the Framingham Heart Study due to its availability to the public and its frequent usage in the medical research as well as machine learning studies on the prediction of cardiovascular disease. It has a wide variety of clinically relevant risk variables and enough data to train and test many classification algorithms.

Furthermore, the data set represents a real-world health care concern, which is particularly pertinent to preventative medicine. Early prediction of coronary heart disease before the presentation of symptoms can assist healthcare workers to identify those patients who might benefit from early intervention, lifestyle change and frequent clinical surveillance. As a result, the Framingham dataset serves as an adequate basis for the design and evaluation of predictive machine learning models for this project.

Chapter 5 – Exploratory Data Analysis

5.1 Introduction

Before developing any machine learning model, it is vital to know the structure and quality of the data collection. Exploratory Data Analysis (EDA) is useful in discovering the properties of the data, detecting possible quality problems, understanding linkages between variables, and determining the preprocessing processes necessary before model construction.

In this project EDA was done using python tools like Pandas, NumPy, Matplotlib, Seaborn etc. The study included dataset structure, summary statistics, missing values, duplicate records, variable distributions, outliers, correlations and class imbalance. These studies provide useful insights to inform the preprocessing and model creation processes.

5.2 Dataset Structure

The Framingham Heart Study dataset comprises 4,240 observations and 16 variables including the objective variable (TenYearCHD). Each row is a single research participant and each column is a demographic, behavioral, or clinical trait relevant to cardiovascular health.

The data collection comprises numerical and category variables. Numeric factors are age, cholesterol level, systolic blood pressure, diastolic blood pressure, body mass index (BMI), heart rate, glucose level, and number of cigarettes smoked each day. Binary categorical variables are factors such as sex, smoking, diabetes, hypertension, prior stroke and medicine for blood pressure.

The structure of the data was examined and it was established that the problem is a supervised binary classification job. This is because the target variable can only have two outcomes, i.e., whether the participant acquired coronary heart disease within 10 years or not.

5.3 Summary Statistics

For the purpose of describing the key properties of the numerical variables, descriptive statistics were produced. The distribution of the data was analyzed by means of the mean, standard deviation, minimum, maximum and quartiles.

Summary Statistics of various Clinical parameters Summary statistics revealed a large variability for various clinical parameters. For example, systolic blood pressure, total cholesterol, glucose level and body mass index had large ranges, reflecting significant variation in cardiovascular risk among participants. Age was also somewhat diverse, suggesting that people of varying ages were included in the sample.

The descriptive statistics give an initial picture of the data set before further analysis was done.

5.4 Missing Value Analysis

Missing values were identified in several variables, including education, cigsPerDay, BPMeds, total cholesterol, BMI, heart rate, and glucose. As the proportion of missing data was relatively low, median imputation was applied because it is robust to outliers and preserves the dataset size.

5.5 Duplicate Record Analysis

Duplicate observations can bias machine learning models by over-representing specific persons or trends. So, we checked duplicate records before preprocessing. The study showed that there were no duplicate rows in the data set. At this moment there were thus no deleted records.

This conclusion indicates that each participant occurred just once in the data set.

5.6 Distribution of Quantitative Variable

The distributions of the numerical variables were investigated in histograms. The knowledge of these distributions helps to know if the variables resemble normal distributions or whether they have skewed values.

Age, systolic blood pressure and body mass index were generally continuously distributed, but glucose and smokes per day were positively skewed with pronounced skewness. Such heterogeneity is expected in clinical datasets as physiological parameters vary significantly from person to person.

Although several variables were not normally distributed, this did not limit the use of machine learning methods, as most of the selected models do not require normally distributed predictors.

5.7 Outlier detection

Boxplots were used to explore outliers in each numerical variable.

Some variables including systolic blood pressure, diastolic blood pressure, cholesterol, BMI, glucose and heart rate have readings outside the normal range. These values are for persons with extremely high or low clinical measures.

These were not eliminated because of the medical plausibility of these observations and the possibility that they reflect people at increased cardiovascular risk. Ignoring these data may impair the model's capacity to detect clinically relevant instances.

In preprocessing, instead of removing outliers, feature scaling was performed to lessen the effect of big numerical discrepancies across variables.

5.8 Correlation Analysis

The correlations between numerical variables were explored using the correlation heatmap.

Most of the variables were weak to moderate correlated, suggesting that the data set is not severely impacted by multicollinearity. Some of the predicted associations were found: positive correlations between systolic and diastolic blood pressure, and between age and many cardiovascular risk variables.

The lack of very strong correlations shows that each variable adds distinct information to the prediction models.

5.9 Classes Distribution

There was an obvious imbalance in the class of the goal variable (TenYearCHD). About 85% of individuals didn't acquire coronary heart disease, but about 15% did develop the condition within 10 years. Such bias may result in machine learning algorithms favoring the prediction of the majority class, leading to poor identification of high-risk patients.

To solve this problem, we used Synthetic Minority Oversampling Technique (SMOTE) exclusively on the training dataset after the train-test split. This strategy balanced the training data and prevented leaking of information into the testing dataset.

5.10 Exploratory Data Analysis Summary

The exploratory data analysis yielded a few key insights that helped us construct the machine learning pipeline:

- The dataset was composed of 4,240 patient records and 16 variables. Some of the predictor variables had missing values which were handled via median imputation.
- No duplicate records were found. Outliers were found in numerous clinical measures, but were preserved as they are actual observations from patients.
- Large class imbalance in the target variable, SMOTE was used before training the model. The correlation study did not provide any evidence of significant

multicollinearity. The dataset has enough demographic and clinical information to be able to create predictive machine learning models.

The exploratory study as a whole indicated that the dataset was adequate for machine learning after the necessary preparation procedures were performed. The findings in this step directly influenced the data preparation techniques detailed in the next chapter.

Chapter 6 – Pre-processing Data

6.1 Introduction

After studying the dataset, many preprocessing procedures were done before training the machine learning models. Healthcare datasets are commonly plagued by missing values, variables on multiple scales, and uneven class distributions, which will degrade the performance of models if not handled. Hence, data preprocessing was done to enhance the quality of data and the machine learning models were trained using the same and trustworthy input data.

The preprocessing pipeline for this project consists of train test split, missing value imputation, feature scaling, addressing class imbalance using SMOTE and feature selection using Recursive Feature Elimination (RFE) Each step was done in a certain order to avoid leaking of knowledge and keep the testing data set totally independent during model assessment.

6.2 Train Test Split

Before any preprocessing activities, the dataset was separated into training and testing subsets. Model training was conducted on approx. 80% of the data, with the remaining 20% utilized for testing. We utilized a random state of 42 for reproducibility of the results.

The split was made before imputation, scaling, and oversampling to avoid leakage of information from the test set into the training process. This is a common machine learning strategy that offers a more robust prediction of how well the model will perform on unseen data.

Moreover, the split was done using “stratified sampling” to maintain the original distribution of classes of the target variable in both training and testing data sets. This avoided the problem of unequal distribution of positive/negative instances of coronary heart disease throughout the two halves of the data set.

6.3 Handling Missing Data

Exploratory Data Analysis The exploratory data analysis revealed that certain variables had missing values: education, smokes per day, blood pressure medicine, total cholesterol, body mass index, heart rate, and glucose.

Instead of deleting these data, “median imputation” was employed to impute missing values. Median was used because it is less affected by extreme values than the mean and is a more robust estimate for clinical data where outliers are typically present.

The imputer was trained on the training dataset just to avoid information leaking. The same fitted imputer was then used for training and testing datasets. This guaranteed that no information from the testing data was used in the model construction.

6.4 Feature Scaling

The Framingham dataset used a variety of units and number ranges for its predictor variables. For example, age is measured in years, blood pressure in millimeters of mercury (mmHg), cholesterol in milligrams per deciliter (mg/dL) and glucose in blood concentration units. These variations might impact distance or gradient based optimization in machine learning systems.

To guarantee that all the variables contribute equally during model training, “StandardScaler” was used to normalize the numerical variables. This change resulted in each variable having a mean of 0 and a standard deviation of 1.

The scaler was fitted exclusively on the training data and then used to transform the testing data using the same transformation parameters. This technique assured the independence of the testing data and avoided information leaking.

6.5 Dealing with Class Imbalance Using SMOTE

One of the main issues found in the exploratory data analysis was the target variable imbalance. 85% of the individuals did not acquire coronary heart disease, whereas only 15% developed illness in 10 years.

Training a model on such uneven data might result in biased predictions since the algorithm may learn to favour the majority class and fail to detect those individuals who are genuinely at danger.

This problem was solved using the Synthetic Minority Oversampling Technique (SMOTE) on the training dataset following feature scaling. SMOTE creates synthetic instances of the minority class, rather than merely copying existing observations. This method balances the training dataset of the model which increases its capacity to identify patients at high risk.

Importantly, SMOTE was applied only to the training data. The same testing dataset was used to guarantee that the model performance was tested on genuine observations that the model had not before seen.

6.6 Feature Selection

The data set comprises several predictor factors, although certain variables are more useful for predicting coronary heart disease than others. The inclusion of unneeded variables can lead to increased model complexity and decreased interpretability.

To choose the most informative predictors, we used Recursive Feature Elimination (RFE) using Logistic Regression as the base estimator. RFE repeatedly eliminates the least important variable until the target amount of features is reached.

The final feature set consisted of ten predictor variables and were utilized to train all three machine learning models. To simplify the models, the number of predictors was reduced while preserving the predictive performance.

6.7 The preprocessing pipeline consisted of the following steps:

1. Load the Framingham Heart Study dataset.
2. Separate predictor variables and the target variable.
3. Split the dataset into training and testing subsets.
4. Perform median imputation on missing values.
5. Standardize numerical variables using StandardScaler.
6. Apply SMOTE to the training dataset only.
7. Perform feature selection using Recursive Feature Elimination (RFE).
8. Train and evaluate the machine learning models.

6.8 Summary

The preprocessing step changed the raw Framingham information into a clean and organized dataset that is appropriate for machine learning. Median imputation was used to deal with the missing values, numerical variables were normalized, SMOTE was used to deal with the class imbalance and RFE was used to choose the most informative predictors. Also, strict efforts were made to avoid information leaking by fitting preprocessing algorithms exclusively on the training data and then applying them to the testing data.

These preprocessing techniques provided a solid base for model construction and served to increase the validity of the predicted results described in the next chapter.

Chapter 7 – Machine Learning Models

7.1 Introduction

We developed three supervised machine learning models to predict 10-year coronary heart disease risk after preprocessing. These models were chosen due to their common use for binary classification problems and their different ways of prediction. We used Logistic Regression as a baseline model and Random Forest and XGBoost to investigate if more complicated algorithms may improve the prediction performance.

7.2. Logistic Regression

Our baseline model was Logistic Regression, chosen for its simplicity, interpretability, and frequent usage in healthcare research. The predictor variables chosen are utilized in the model to estimate the probability of getting coronary heart disease. Then the model was trained on the training data with SMOTE applied.

Benefits:

- Easy to read.
- Quick learner.
- Commonly used for medical prediction studies.

7.3 Random Forest

Random Forest is an ensemble learning technique. It works by constructing several decision trees and then using them to improve forecast accuracy and control over-fitting. It can learn complex correlations between variables. It also provides feature relevance ratings.

Advantages:

- Can handle nonlinear relationships.
- Reduces chance of over-fitting.
- Provides feature importance.

7.4 XGBoost

XGBoost is a gradient boosting decision tree approach that is developed incrementally to improve the performance of the forecasts. It is very accurate and efficient, and is widely used in healthcare and other predictive modeling applications.

Benefits:

- Good predictive performance
- Deals with complex data patterns.
- Built-in regularization.

7.5 Metrics for Evaluation

All the three models were tested on the five performance indicators as:

- Accuracy – Percentage of correct responses predicted overall.
- Precision – Of all the predicted positive cases, how many were actually positive.
- Recall - % of true positives that were discovered correctly.
- F1-score– compromise between accuracy and recall
- ROC-AUC– The overall ability of the model to differentiate between the positive and negative scenarios.

Accuracy alone is not a reliable metric as health care data is unequal, hence other assessment methods are required.

Chapter 8 - First Results and Evaluation of the Model

8.1 Introduction

Following data pre-processing and model creation, three machine learning models were assessed for the prediction of 10-year risk of coronary heart disease. To compare fairly, we ran Logistic Regression, Random Forest and XGBoost with the identical testing dataset.

The model performance was evaluated using many measures such as Accuracy, Precision, Recall, F1-score and ROC-AUC. Multiple assessment measures are used to better identify the strengths and limitations of each model, especially as the target variable in the dataset is skewed.

8.2 Accuracy of the Model

The performance of the three machine learning models is shown in table 1.

The data indicate that each model behaved differently. XGBoost had the best overall accuracy (75.4%) suggesting that it accurately categorized the most patients. But accuracy isn't the full picture of model performance, especially in healthcare applications where the cost of missing a patient with heart disease might be high.

Logistic Regression has the greatest Recall (58.1%), F1-score (0.343) and ROC-AUC (0.694). What this means is that the model was more accurate in picking out people who were genuinely at risk of getting coronary heart disease. Random Forest showed an intermediate performance but did not surpass the other two models in any of the assessment measures

8.3 Matrix Analysis of Confusion

We built confusion matrices for all three models to observe how the predictions were divided across the positive and negative classes.

The confusion matrices demonstrate that a high number of people who did not acquire coronary heart disease were properly recognized by all three models. But the models vary in their capacity to detect people who actually get the condition. Logistic Regression had more True Positive predictions than the other models which resulted in a better Recall score.

These data indicate that although XGBoost had greater overall accuracy, Logistic Regression was better at identifying high-risk patients.

8.4 Receiver Operating Characteristic Curve (ROC)

The Receiver Operating Characteristic (ROC) curve was used to assess the performance of each model to discriminate between individuals who acquired coronary heart disease and those who did not.

From the ROC curves, the model with the highest Area Under the Curve (ROC-AUC) is Logistic Regression (ROC-AUC = 0.694), followed by Random Forest (0.653) and XGBoost (0.650) respectively. The greater the ROC-AUC the better the discriminating between positive and negative instances across different categorization thresholds.

The differences amongst the models were rather slight, although Logistic Regression was consistently better at discriminating than the two ensemble learning models.

8.5 Model Comparison

Comparing the three machine learning models it seems that each algorithm has distinct strengths.

XGBoost provided the best accuracy, which indicates that this classifier did well in the aspect of overall classification. However, its Recall was lower than Logistic Regression, which means that it missed a bigger proportion of individuals that really got coronary heart disease.

Logistic Regression has the best Recall, F1-score and ROC-AUC score. Here, because the research is about identifying people at risk of heart disease, Recall is a crucial metric as it calculates how many of the true positive instances are accurately detected. A missed patient with true risk postpones preventive therapy and increases the possibility of cardiovascular problems in the future.

Random Forest gave balanced performance but did not rank first in any assessment criteria. Nonetheless, it still has its use since it provides good model interpretability by feature significance analysis.

Table 1.

	Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
0	Logistic Regression	0.662	0.244	0.581	0.343	0.694
1	Random Forest	0.703	0.234	0.419	0.300	0.653
2	XGBoost	0.754	0.262	0.341	0.296	0.650

8.6 Preliminary Findings

Early results suggest that the 10-year risk of coronary heart disease may be predicted with fair accuracy by demographic, behavioral and clinical characteristics.

Among the 3 models tested, Logistic Regression provided the best overall performance for healthcare screening with the greatest Recall and ROC-AUC. While XGBoost has the best Accuracy, its lower Recall indicates that it might not be able to catch more high-risk patients.

The results suggest that the best machine learning model should not be selected only on a single assessment criterion. Instead, certain performance criteria should be examined to guarantee that the selected model achieves the healthcare application objectives.

Chapter 9: Explaining Your Model Using SHAP

9.1 Introduction

Machine learning models can produce accurate predictions, but typically do not give explanations of how different factors contribute to the predictions. In healthcare, knowing why a model predicts a patient is at high risk is almost as crucial as the forecast itself. Thus, to enhance the model Explainability, this study used SHAP (SHapley Additive explanations).

SHAP is an explainable artificial intelligence (XAI) method estimating the contribution of each attribute to the model's predictions. It allows for the identification of the most important clinical and demographic factors for predicting coronary heart disease and increases the interpretability of machine learning models.

9.2 SHAP Summary Analysis

The SHAP values were generated to evaluate the relevance of each specified attribute. The SHAP summary graphic visualizes the effect of each predictor on the model for all data. The features with higher SHAP values contributed more to the prediction of coronary heart disease.

The SHAP summary figure shows that various variables are regularly affecting the model predictions. The size and direction of the SHAP values reflect how the increase or decrease in the anticipated risk of coronary heart disease is affected by each patient attribute.

9.3 Feature importance

The SHAP feature importance analysis found the variables that had the strongest contribution to the predictions of the model.

Analysis revealed that “age” was the strongest predictor of coronary heart disease. This result is in line with what is already known in medicine, since the risk for cardiovascular disease grows with age.

Other major factors were:

Age, Systolic Blood Pressure (sys BP), daily number of cigarettes, Male Sex, Blood pressure (diastolic), Cholesterol, Total Heart rate, Blood sugar, Blood Pressure Control Medication, Prior stroke.

The other predictors were less important in the model than these variables and had a significant impact in deciding the final results.

9.4 Results Interpretation

The SHAP study suggests that the machine learning model included clinically significant cardiovascular risk variables as the main reasons for its predictions. The best indicators were age and blood pressure readings, followed by smoking habit and cholesterol level. These findings are consistent with the known clinical evidence that these factors are substantial contributors to coronary heart disease.

The agreement of the machine learning model with known cardiovascular risk variables enhances the trust in reliability and therapeutic usefulness of the created prediction model. This also implies that the model is learning significant patterns from the data rather than random connections.

9.5 Recap

The SHAP analysis made the machine learning model more interpretable by showing the factors that had the most impact on predicting coronary heart disease. The most important indicators were age, systolic blood pressure, cigarette usage, cholesterol level, glucose and sex. These results are consistent with previous cardiovascular studies and highlight the need for explainable machine learning approaches in healthcare.

The data reported in this chapter contribute to answering the third research question, i.e., what demographic and clinical characteristics have the most impact on the prediction of coronary heart disease.

Discussion - 10th Chapter

The results are discussed in the discussion chapter in relation to the study topics and previous knowledge. This part isn't only about stating performance indicators, but about interpreting the results and their significance.

The initial study goal was to evaluate whether demographic and clinical characteristics might be utilized to predict the ten-year risk of coronary heart disease. Results demonstrated that all three machine learning models can identify individuals at risk, showing that characteristics such as age, blood pressure, cholesterol, smoking habit, glucose level and body mass index had useful predictive information.

The second study topic was on the performance of Logistic Regression, Random Forest, and XGBoost. Overall, XGBoost was the best performing with an accuracy of 75.4%, while Logistic Regression had the greatest Recall (58.1%), F1-score (0.343) and ROC-AUC (0.694). Higher accuracy is often desired but not necessarily the suitable metric for healthcare applications. In illness prediction, precisely identifying those individuals who are truly at risk is sometimes more essential than optimizing overall accuracy. Thus, Logistic Regression was better for this particular healthcare problem since it identified a higher number of high-risk individuals.

The third study aim was to identify the factors that were strongest contributors to model predictions. SHAP analysis indicated that the most important variables were age, systolic blood pressure, cigarette intake, cholesterol, glucose level and sex. These findings are consistent with the well-established cardiovascular literature, indicating that the machine learning model successfully learnt clinically relevant associations from the data set.

Overall, the results imply that machine learning can assist in early identification of those at higher risk of coronary heart disease. But the results also suggest that for model selection, rather than just accuracy, one should consider many assessment measures. Recall and ROC-AUC are also useful to healthcare prediction as they tell us how well the model is able to identify individuals that could require further medical treatment.

Chapter 11 – Limitations

There are certain limitations of this study that should be mentioned. First, the study used data only from the Framingham Heart Study, and hence the results may not be applicable to other populations or healthcare systems. Second, we applied median imputation and SMOTE to deal with missing values and class imbalance, however these approaches may still introduce some bias.

Furthermore, the models were trained and evaluated on only one train-test split, not cross-validation that would give a more rigorous evaluation of the performance. We have only reviewed three machine learning methods and future research may test more models to improve the prediction accuracy. These models are ultimately intended to be a tool for health care providers—not a substitute for clinical judgment. “While these limitations exist, this study shows that machine learning can be a helpful tool to identify patients at higher risk of heart disease.

Chapter 12 – Conclusion

I constructed and tested machine learning models to predict 10-year risk of coronary heart disease from the Framingham Heart Study dataset. The workflow involved data pre-processing, feature selection, model training, assessment and interpretation.

Logistic Regression is the highest performing model overall for identifying high risk patients among the three models examined (Logistic Regression, Random Forest and XGBoost) and so is the most appropriate model for this study. SHAP study found that age, systolic blood pressure, smoking, cholesterol, glucose and sex were the most significant predictors of heart disease.

Overall, the results suggest that machine learning might be a useful technique in helping early diagnosis of coronary heart disease. With additional validation on bigger and more diversified datasets these models might provide useful support for clinical decision-making.

AI Productivity Declaration

The ChatGPT was solely used as an additional learning tool for explaining ideas. It was not used to develop the results or conclusions of the research. All data pre-processing, model training, evaluation, interpretation and final judgments were done separately.

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